AI-Powered Resume Analyzer and Career Recommendation System

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Introduction

The project is an intelligent web-based platform that helps job seekers discover suitable career paths and strengthen their professional profiles. It analyzes uploaded resumes using NLP and ML, predicts job domains through a BERT-based model, and recommends matching job opportunities and relevant certification courses. The system also generates a resume strength score based on education, skills, certifications, and experience. Built with a Flask backend and an interactive web interface, it delivers real-time insights to help users assess their profiles and enhance employability.

Problem Statement

- Job seekers struggle to match their resumes with suitable career opportunities.
- Many apply blindly without knowing job fit or skill gaps.
- Manual job searches take excessive time and effort.
- No automated feedback exists to improve resume quality.
- Career and learning resources are scattered across multiple platforms.

Proposed Solution

- An AI-powered system that analyzes resumes using NLP and machine learning.
- Predicts job domains and provides personalized job recommendations.
- Generates resume scores based on education, skills, experience, and certifications.
- Suggests relevant courses to close skill gaps and boost employability.
- Saves time with real-time, data-driven insights through a unified web platform.

Methodology

The system is developed using a dual-layer architecture that integrates machine learning pipelines with a web-based frontend and backend framework.

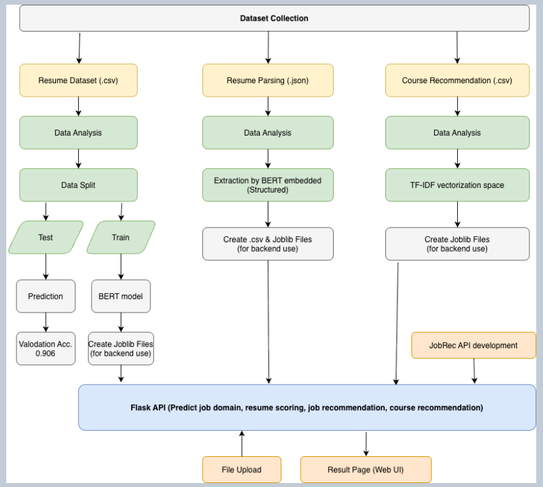


Fig.: Architecture of Backend AI Pipeline

At the backend, three primary datasets are used, and each dataset undergoes analysis and feature extraction to transform unstructured text into meaningful data. The backend employs NLP and ML models for classification, scoring, and recommendation tasks. A BERT model is utilized to identify job roles from resumes, while TF-IDF vectorization supports job and course recommendation modules. All trained components are stored as .joblib files for efficient backend use and rapid prediction within the API.

The frontend is built using HTML, CSS, and JS, offering a responsive and user-friendly interface where users can upload resumes, create accounts, and view intelligent recommendations in real time. The Flask server functions as a bridge between the AI models and

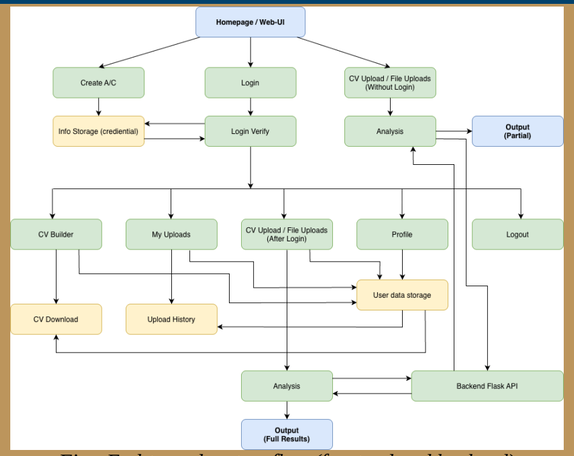


Fig.: End-to-end system flow (frontend and backend)

the web interface, user authentication, model inference, and result visualization.

This integrated workflow transforms uploaded resumes into structured insights, including predicted job domains, resume strength, job scope, and course recommendations, forming a unified AI-driven career guidance platform.

Result and Analysis

The system demonstrated stable and efficient learning behavior during model training and deployment.

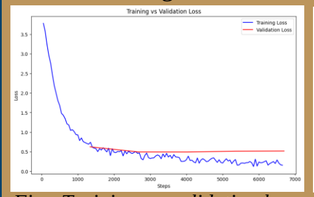


Fig.: Training vs. validation loss



Fig.: validation accuracy

The loss curve shows a sharp decline in training loss to below 0.3, while the validation loss stabilized around 0.5, indicating strong model generalization. The validation Acc. curve shows 90.6% in predicting job domains.

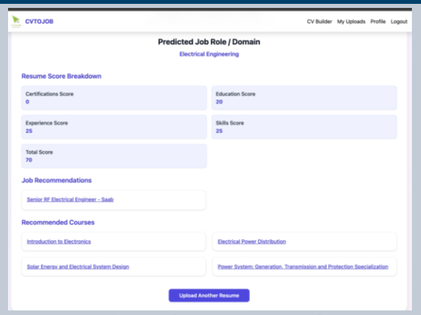


Fig.: Web-UI output result

The web-UI interface presents results in real time, which predict a job domain with a resume score across education, skills, experience, and certifications. It also provides job recommendations and certification courses to help users upskill and enhance employability.

Conclusion

The project delivers an AI-powered platform that automates resume analysis, predicts job domains, scores resumes, matches jobs, and recommends courses. It provides data-driven career insights, helping job seekers for suitable opportunities, and enhances employability through intelligent, personalized guidance.

Key References

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